

King Mongkut's Institute of Technology Ladkrabang
School of Engineering – Robotics and AI Engineering



01416318 MICROPROCESSOR AND INTERFACE

Line following Report_RAI_Group16

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Date of Submission : 29/05/2023

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Introduction

1. Line-following robots are autonomous systems that navigate along a predefined path by detecting and tracking lines on a surface.
2. This report presents the design and implementation of a line-following robot, focusing on the utilization of STM32 microcontroller, along with the implementation of Proportional-Integral-Derivative (PID) control and Pulse Width Modulation (PWM) techniques.
3. The report discusses the design of a Printed Circuit Board (PCB) for integrating the components of the line-following robot.

Purpose

1. To discuss the physical structure, chassis design, sensor placement, and overall system architecture.
2. To make the integration and layout of the electronic components on the Printed Circuit Board (PCB).
3. To implement Proportional-Integral-Derivative (PID) control and Pulse Width Modulation (PWM) techniques in the robot's operation.

Scope

1. To be capable of following a line of a predefined path, detecting and tracking the line within 3 minutes.
2. To allow the calibration of the line thickness threshold.
3. To take various degrees of turns along the predefined path. It should be able to navigate both sharp and gradual curves smoothly and efficiently.

Methodology

1. Block Diagram

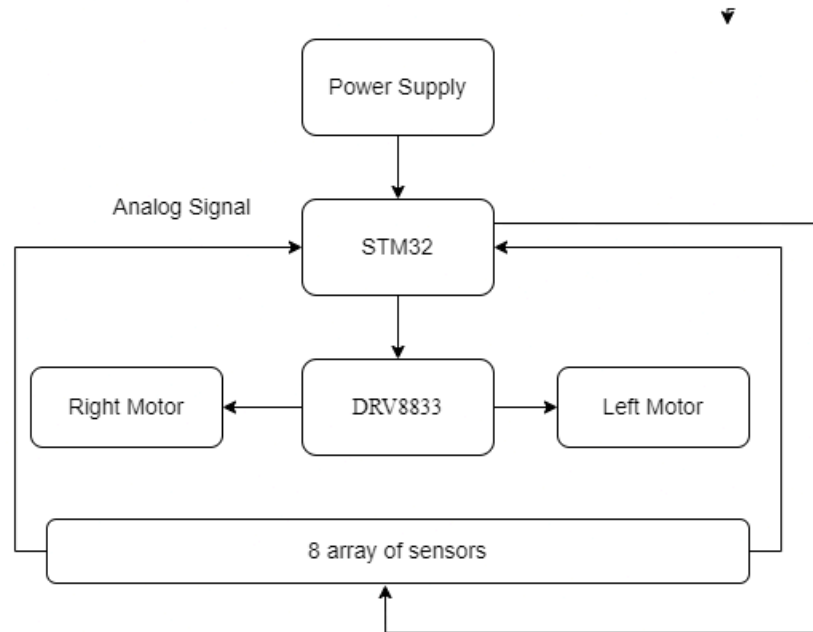


Figure of Block Diagram

2. Hardware Required

2.1 Microcontroller STM32 F401RE

The STM32 F411RE microcontroller provides a wide range of features that are essential for the line-following robot's functionality. It offers multiple digital and analog I/O pins, allowing for the connection and control of various components and sensors. The microcontroller also includes built-in peripherals such as UART, and PWM, enabling ADC/DMA communication and interfacing with other devices.

2.2 Optical Sensors and the Arrangement

In the design of the line-following robot, a total of 8 sensors are utilized, which are arranged in a curved pattern underneath the front of the robot. These sensors are positioned in

close proximity to the surface to accurately detect the line. The distance between two adjacent sensors is set at 10 mm, allowing for precise line tracking.

To obtain data from the sensors, an analog signal is utilized. The sensors work by emitting light and measuring the amount of light reflected back from the surface. Based on the intensity of the reflected light, the sensors generate an analog signal that provides information about the position of the line relative to each sensor.

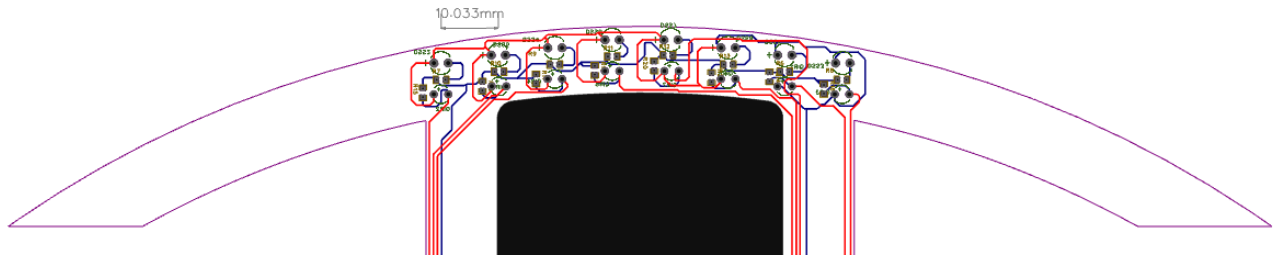


Figure of sensor arrangement

2.3 Motor Driver DRV8833 and Motors x 2

With the DRV8833 motor driver, the line-following robot can achieve precise motor control. It supports PWM (Pulse Width Modulation) control, allowing for speed and direction control of the motors.

The integration of the DRV8833 motor driver and the two motors enables the line-following robot to navigate along the predefined path. The motor driver provides the necessary control and power distribution, while the motors translate that control into physical motion, allowing the robot to accurately follow the line.

2.4 LED and Switch x 2

1. Sensor Calibration Switch and LED:

The sensor calibration switch is used to initiate the calibration process. When the switch is activated, the robot enters the calibration mode, and the corresponding LED associated with sensor calibration lights up.

2. Sensor Reset Switch and LED :

The sensor reset switch is dedicated to resetting the sensors. When the switch is triggered, the robot performs a reset of the sensors, and the corresponding LED associated with

sensor reset lights up or changes its state. This LED provides visual feedback to the user, indicating that the sensor reset process is underway.

2.5 Li-Po battery - 11.1V 550mAh

For supplying power to the STM32 microcontroller and the DRV8833 motor driver along with the two motors in the line-following robot, an 11.1V 550mAh LiPo battery is used. The capacity of 550mAh provides a sufficient runtime for the line-following robot, balancing power needs and weight considerations.

The 11.1V voltage rating of the LiPo battery matches the power requirements of the DRV8833 motor driver, which typically operates at voltages within that range. Additionally, the STM32 microcontroller can handle this voltage range with the appropriate voltage regulation circuitry.

3. Schematic

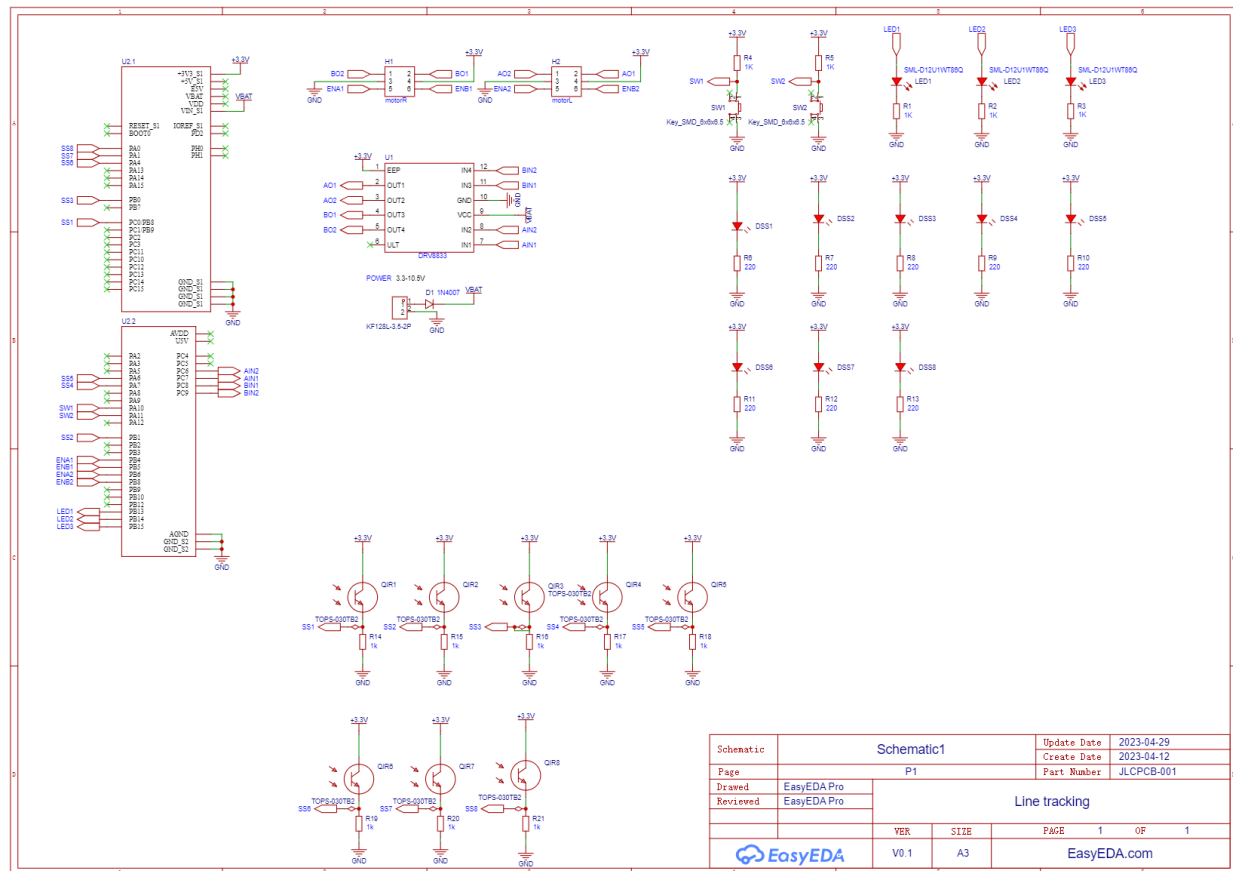


Figure of Circuit diagram

4. PCB Design

By carefully designing the PCB layout, space utilization has been optimized, allowing for a compact design. The placement of components, traces, and vias on the PCB has been strategically planned to minimize signal interference and improve overall system performance.

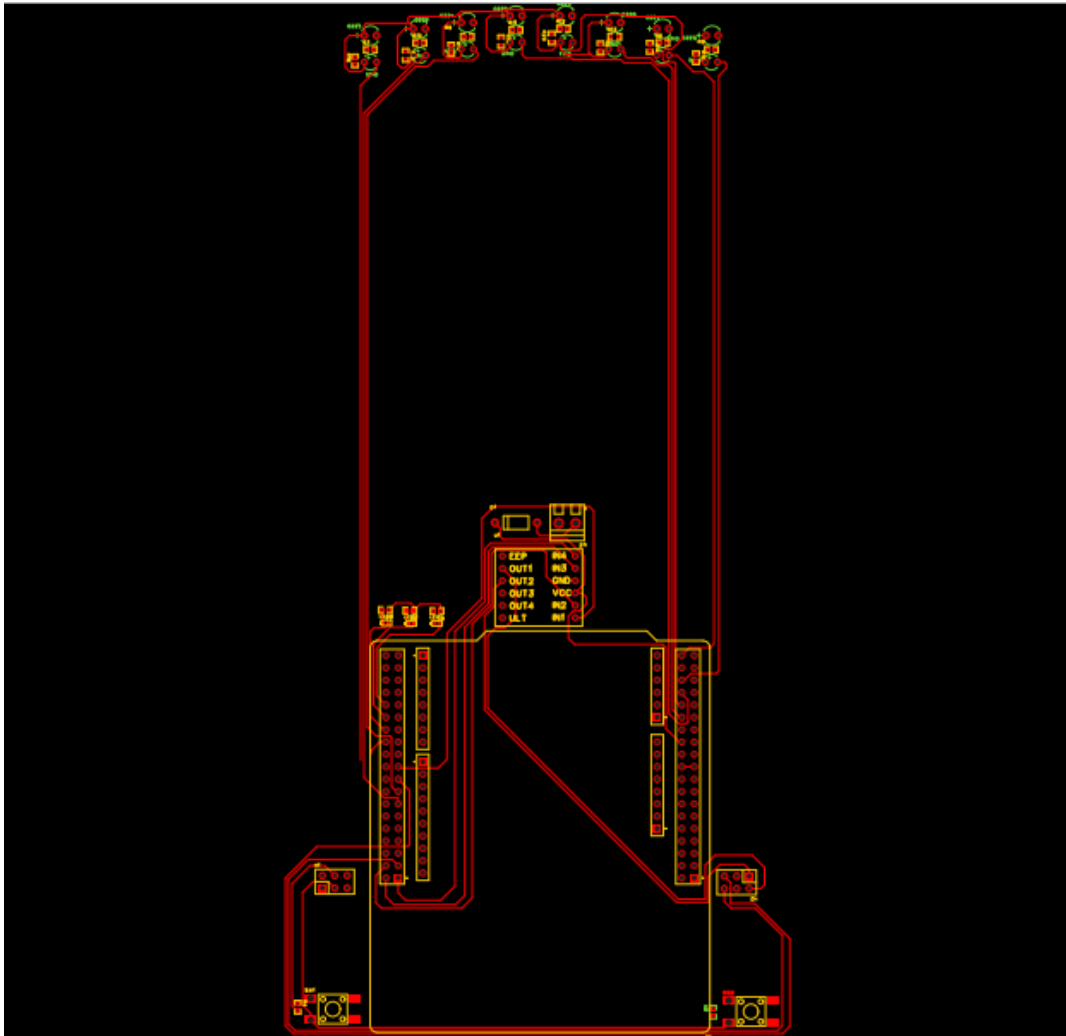


Figure of PCB Design of the upper layer

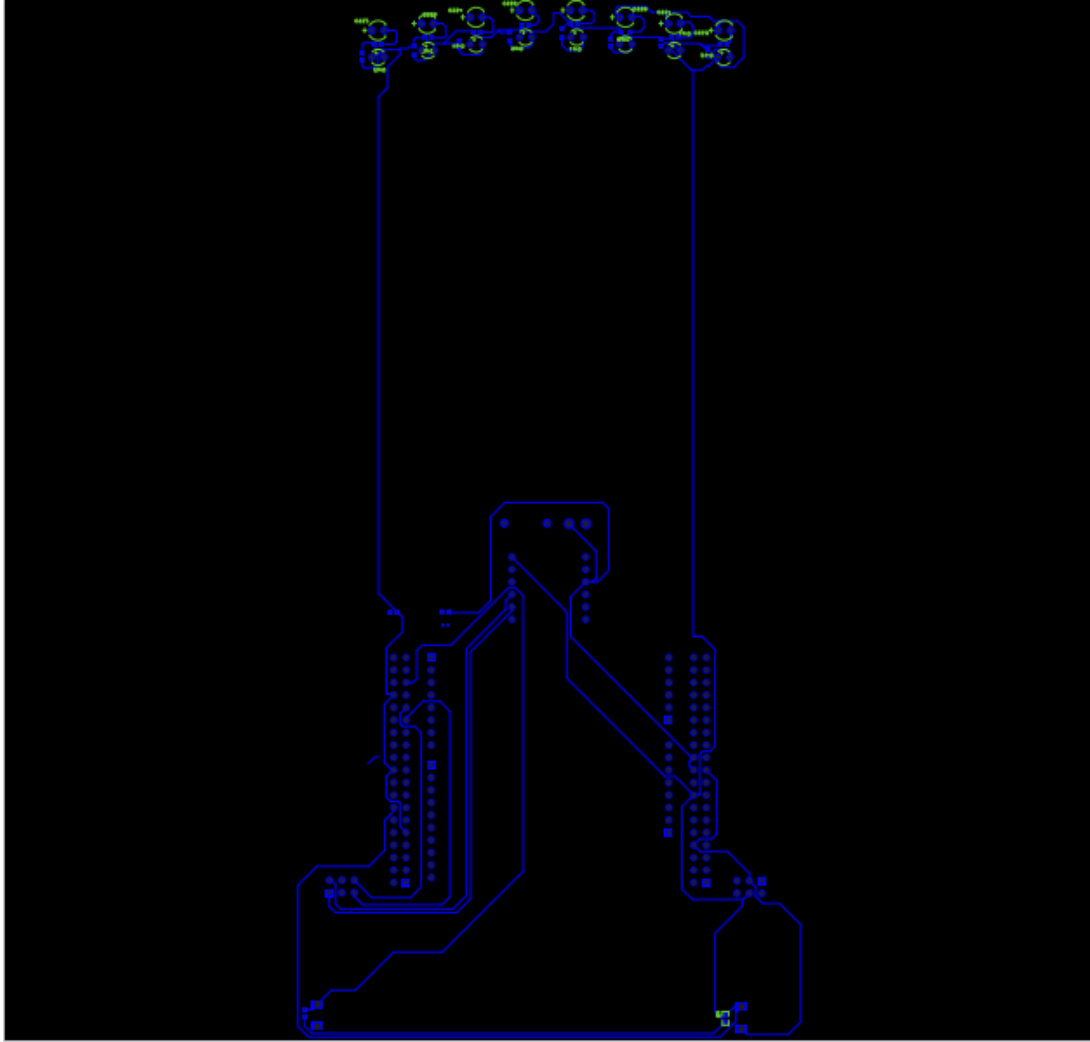


Figure of PCB Design of the lower layer

5. PIN Configuration in STM32

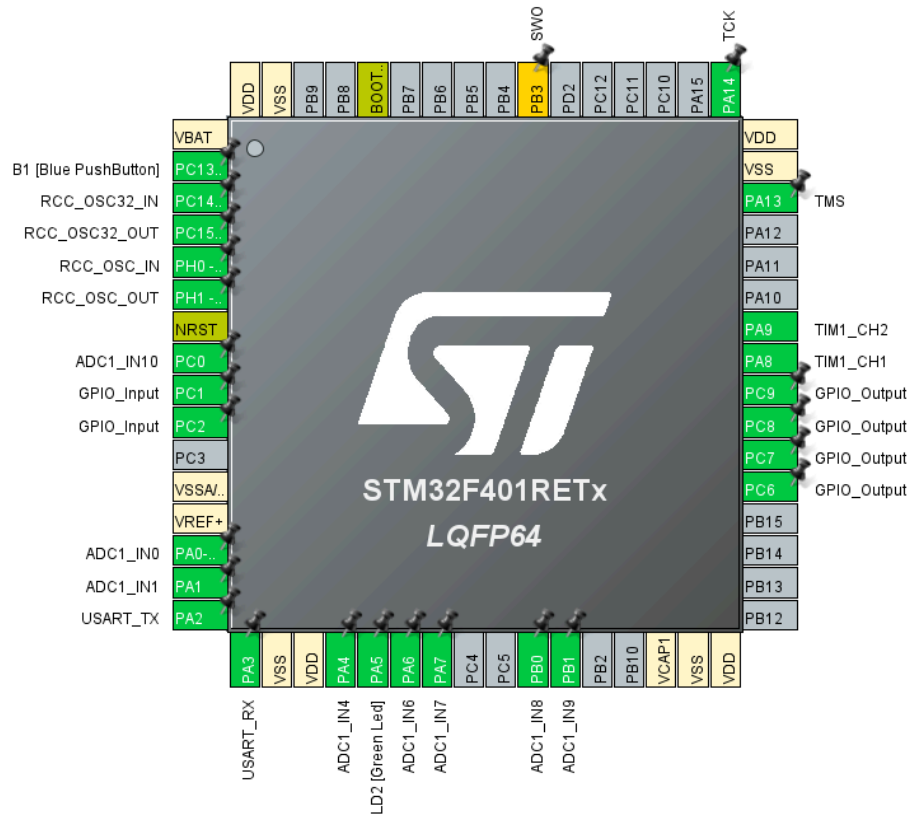


Figure of PIN configuration in STM32

Timer - PWM generator

TIM1CH1 - PA8

TIM1CH2 - PA9

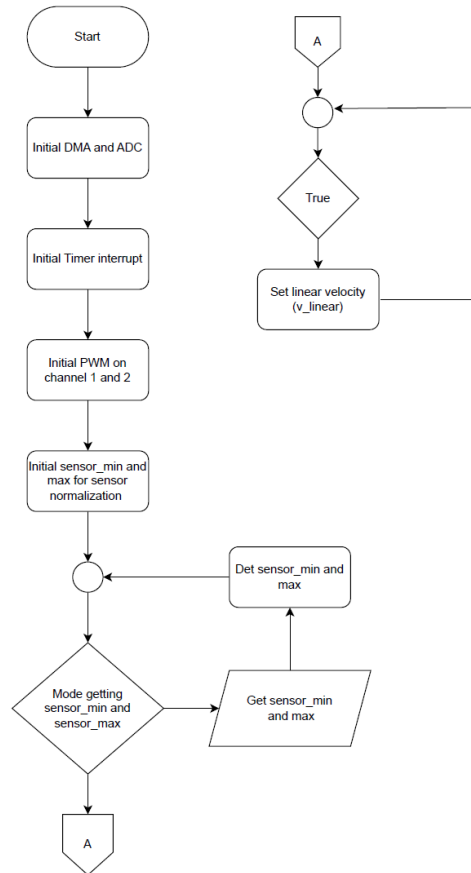
Analog - ADC1 (8 Sensors) - PA0, PA1, PA4, PA6, PA7, PB0, PB1, PC0

GPIO

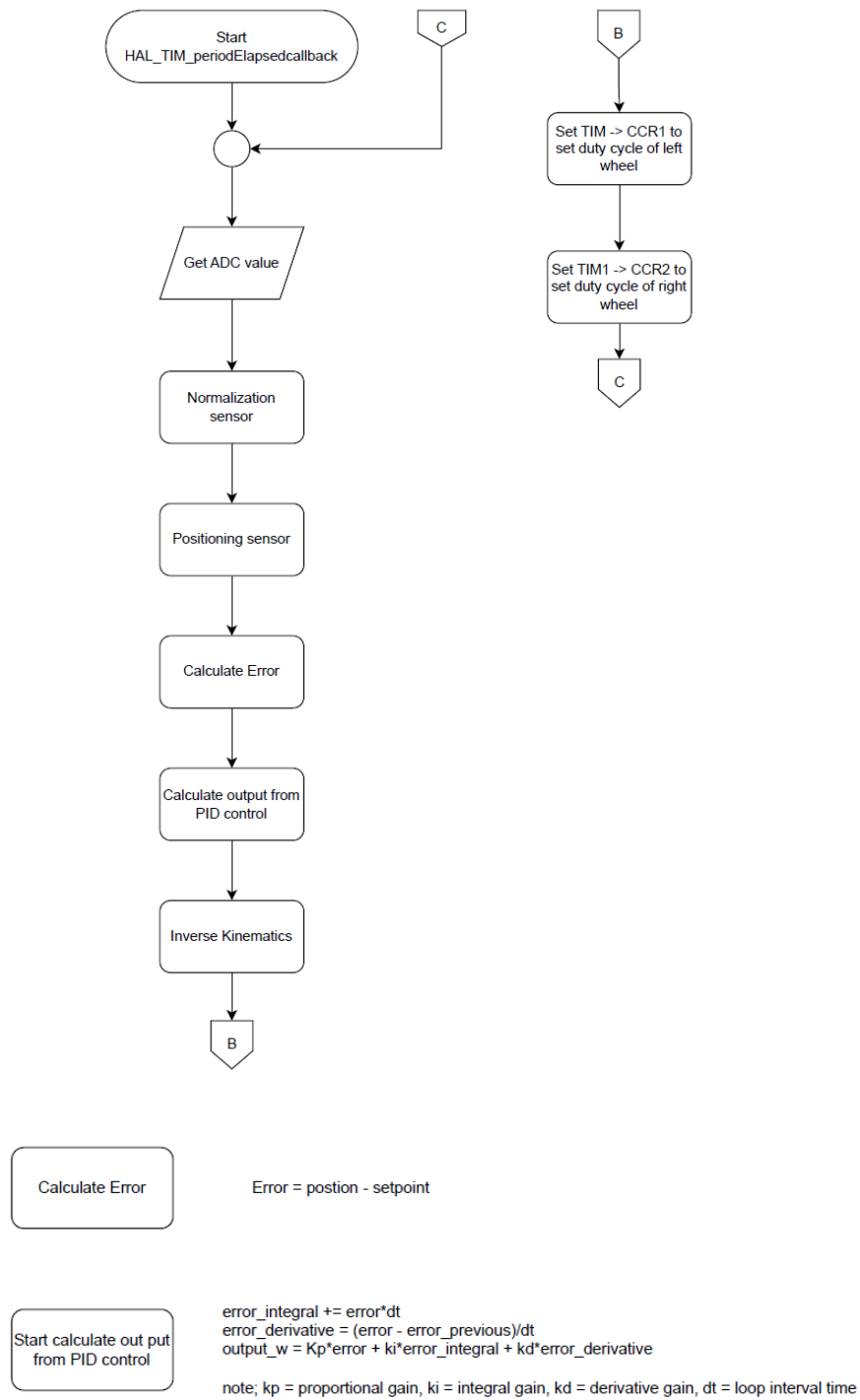
Output (Motor Driver) - PC9, PC 8, PC 7, PC6

6. Flow Chart and Working Logic

6.1 Main program



6.2 Interrupt function



Result

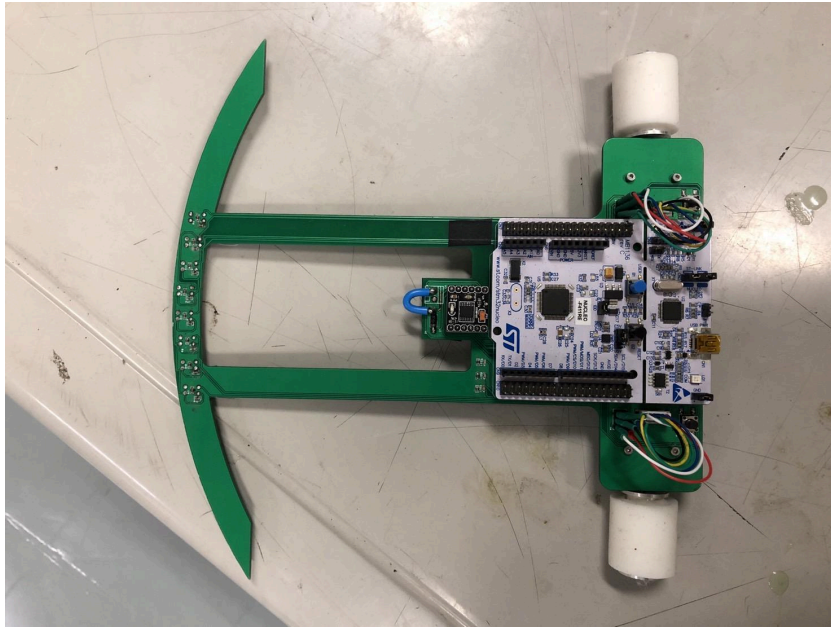


Figure above shows the top view of the speed tracking robot

Time average

Track 1: average time is 10.20 second

Track 2: average time is 50.32 second

Conclusion

In the development of an autonomous line-following robot, the utilization of the STM32 microcontroller has provided efficient control and coordination of the robot's movements, while the implementation of PID control and PWM techniques has enhanced its ability to accurately follow the line and make turns.

Overall, the successful integration of the various components, including the STM32 microcontroller, DRV8833 motor driver, sensors, and PCB, has resulted in the development of an autonomous line-following robot capable of navigating along a predefined path.

Member participation

Hnin Ei San (65011303)

Responsibility: Coding, Design PCB board, Design body of car, soldering

Percentage: 33.33%



(Hnin Ei San)

Phattawin Kummaraphat (65011453)

Responsibility: Coding, Design PCB board, Design body of car, soldering

Percentage: 33.33%



(Phattawin Kummaraphat)

Promptorn Thongpech (65011487)

Responsibility: Coding, Design PCB board, Design body of car, soldering

Percentage: 33.33%



(Promptorn Thongpech)